# HW1_314707047_連鈞宥

2026-03-04

```
library(readxl)
library(PoEdata)
```

```
#data("star5")
#head(star5)
#data("star")
#head(star)
# 找不到資料集star5 & star5_small。only star。
```

```
# 第22題
star5 <- read_excel("C:/Users/tiffa/OneDrive/Desktop/eco_fin/star5.xlsx")
head(star5)
```

```
## # A tibble: 6 × 19
##   absent  aide black   boy freelunch    id mathscore readscore regular  schid
##    <dbl> <dbl> <dbl> <dbl>     <dbl> <dbl>     <dbl>     <dbl>   <dbl>  <dbl>
## 1      5     1     1     1         0 10133       478       427       0 169280
## 2     28     1     0     0         0 10246       494       450       0 218562
## 3      2     0     1     0         1 10263       513       483       0 205492
## 4     10     0     0     1         0 10266       513       456       1 257899
## 5      3     1     0     1         0 10275       468       411       0 161176
## 6      2     0     0     1         0 10281       473       443       0 189382
## # ℹ 9 more variables: schrural <dbl>, schurban <dbl>, small <dbl>,
## #   tchexper <dbl>, tchid <dbl>, tchmasters <dbl>, tchwhite <dbl>,
## #   totalscore <dbl>, white_asian <dbl>
```

```
# a.
#total
star5$totalscore <- star5$mathscore + star5$readscore
model_222 <- lm(totalscore ~ small, data = star5)
summary(model_222)
```

```
##
## Call:
## lm(formula = totalscore ~ small, data = star5)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -283.20  -50.94   -7.20   42.80  334.80
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  918.201      1.155  795.00  < 2e-16 ***
## small         13.741      2.107    6.52 7.61e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 73.48 on 5784 degrees of freedom
## Multiple R-squared:  0.007297,   Adjusted R-squared:  0.007125
## F-statistic: 42.51 on 1 and 5784 DF,  p-value: 7.609e-11
```

```
# b.
# read
model_read <- lm(readscore ~ small, data = star5)
summary(model_read)
```

```
##
## Call:
## lm(formula = readscore ~ small, data = star5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -120.089  -22.089   -4.089   15.911  191.911
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 435.0887     0.4969 875.539  < 2e-16 ***
## small         5.4631     0.9067   6.025 1.79e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 31.62 on 5784 degrees of freedom
## Multiple R-squared:  0.006237,   Adjusted R-squared:  0.006066
## F-statistic:  36.3 on 1 and 5784 DF,  p-value: 1.793e-09
```

```
# math
model_math <- lm(mathscore ~ small, data = star5)
summary(model_math)
```

```
##
## Call:
## lm(formula = mathscore ~ small, data = star5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -163.113  -34.113   -5.113   29.887  142.887
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 483.1126     0.7473 646.468  < 2e-16 ***
## small         8.2775     1.3635   6.071 1.36e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 47.55 on 5784 degrees of freedom
## Multiple R-squared:  0.006331,   Adjusted R-squared:  0.006159
## F-statistic: 36.85 on 1 and 5784 DF,  p-value: 1.355e-09
```

```
# c.
model_aide <- lm(totalscore ~ aide, data = star5)
summary(model_aide)
```

```
##
## Call:
## lm(formula = totalscore ~ aide, data = star5)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -289.50  -51.36   -7.36   42.50  334.64
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  924.497      1.205 767.503  < 2e-16 ***
## aide          -6.140      2.027  -3.029  0.00247 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 73.69 on 5784 degrees of freedom
## Multiple R-squared:  0.001584,   Adjusted R-squared:  0.001411
## F-statistic: 9.174 on 1 and 5784 DF,  p-value: 0.002466
```

```
# d.
model_read_aide <- lm(readscore ~ aide, data = star5)
summary(model_read_aide)
```

```
##
## Call:
## lm(formula = readscore ~ aide, data = star5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -122.435  -21.438   -4.435   15.565  191.562
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 437.4347     0.5182  844.19   <2e-16 ***
## aide         -1.9966     0.8720   -2.29   0.0221 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 31.7 on 5784 degrees of freedom
## Multiple R-squared:  0.0009055,  Adjusted R-squared:  0.0007328
## F-statistic: 5.242 on 1 and 5784 DF,  p-value: 0.02208
```

```
model_math_aide <- lm(mathscore ~ aide, data = star5)
summary(model_math_aide)
```

```
##
## Call:
## lm(formula = mathscore ~ aide, data = star5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -167.062  -33.062   -3.062   30.081  143.081
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  487.062      0.779 625.271  < 2e-16 ***
## aide          -4.143      1.311  -3.161  0.00158 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 47.66 on 5784 degrees of freedom
## Multiple R-squared:  0.001724,   Adjusted R-squared:  0.001551
## F-statistic: 9.989 on 1 and 5784 DF,  p-value: 0.001583
```

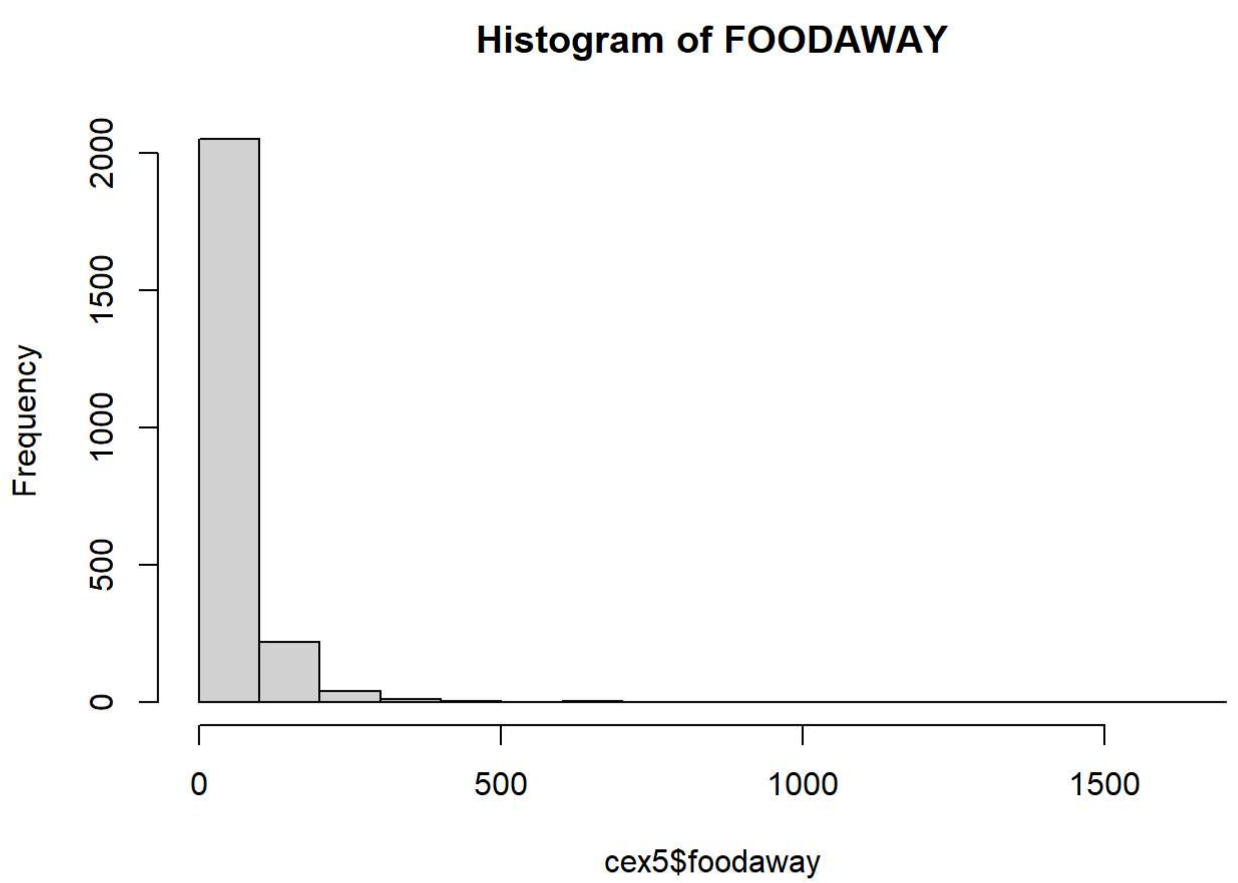
```
# 第25題
cex5 <- read_excel("C:/Users/tiffa/OneDrive/Desktop/eco_fin/cex5.xlsx")
head(cex5)
```

```
## # A tibble: 6 × 24
##       id  year month count occasion advanced alcbev appar college entert  food
##    <dbl> <dbl> <dbl> <dbl>    <dbl>    <dbl>  <dbl> <dbl>   <dbl>  <dbl> <dbl>
## 1 251162  2013     1     1        1        0   6.67  27.8       1   38   108.
## 2 251210  2013     1     1        1        0   1.67 108.        1   20.3 195.
## 3 251214  2013     1     1        1        0   0      0         0   27.2  37.8
## 4 251247  2013     1     1        1        0   0      0         1   31.7 188.
## 5 251271  2013     1     1        1        0  36.7  221.        0  305.   72.2
## 6 251326  2013     1     1        1        0  20     31.4       0  196.  260
## # ℹ 13 more variables: foodaway <dbl>, health <dbl>, hsplus <dbl>,
## #   income <dbl>, midwest <dbl>, nohs <dbl>, northeast <dbl>, older <dbl>,
## #   read <dbl>, rural <dbl>, smsa <dbl>, south <dbl>, west <dbl>
```

```
# a.
summary(cex5$foodaway)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00   14.12   33.15   51.45   69.44 1645.56
```

```
hist(cex5$foodaway, main="Histogram of FOODAWAY")
```



```
# b.
# advanced
summary(cex5$foodaway[cex5$advanced == 1])
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00   21.67   48.34   75.37   89.86 1179.00
```

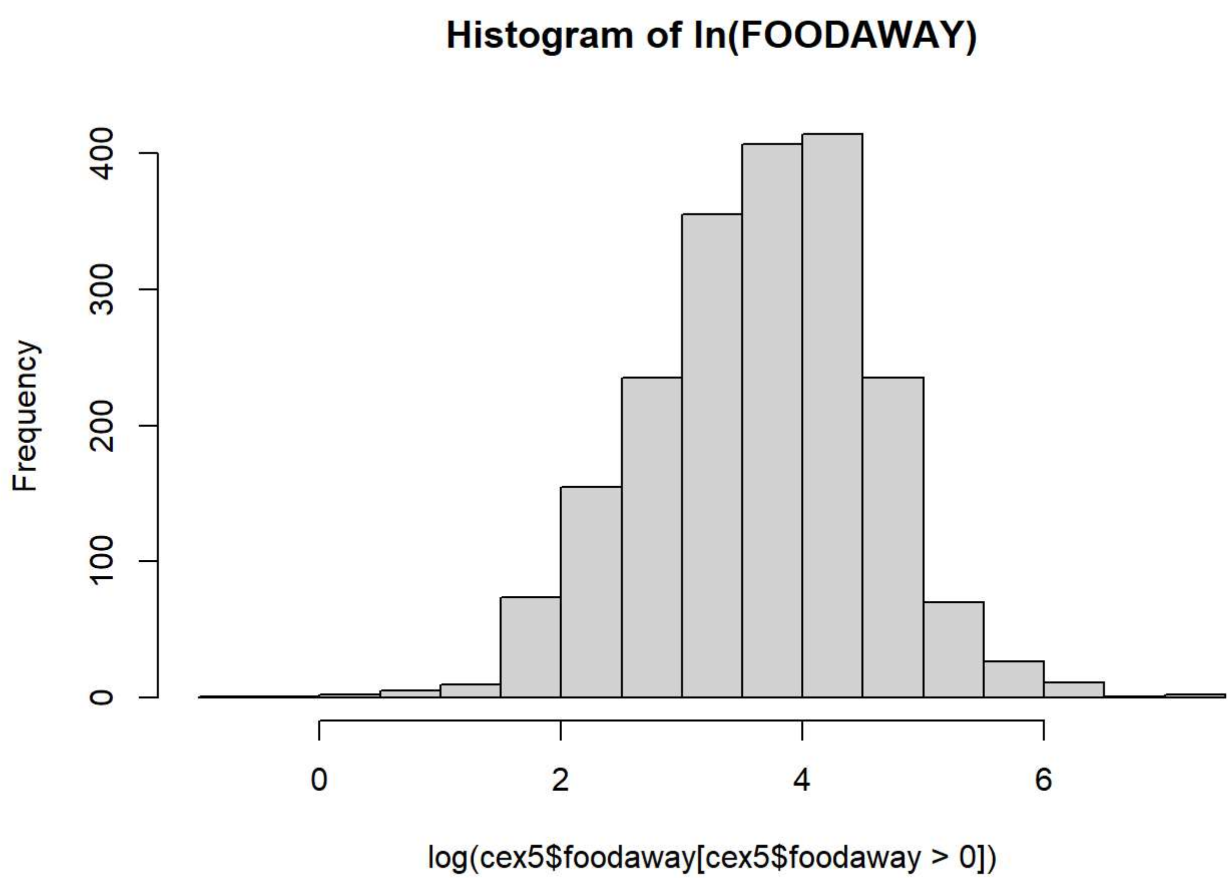
```
# college
summary(cex5$foodaway[cex5$college == 1])
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00   14.90   36.11   54.90   72.22 1645.56
```

```
# no advanced or college
summary(cex5$foodaway[cex5$advanced == 0 & cex5$college == 0])
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00    9.63   25.02   38.32   55.97  437.78
```

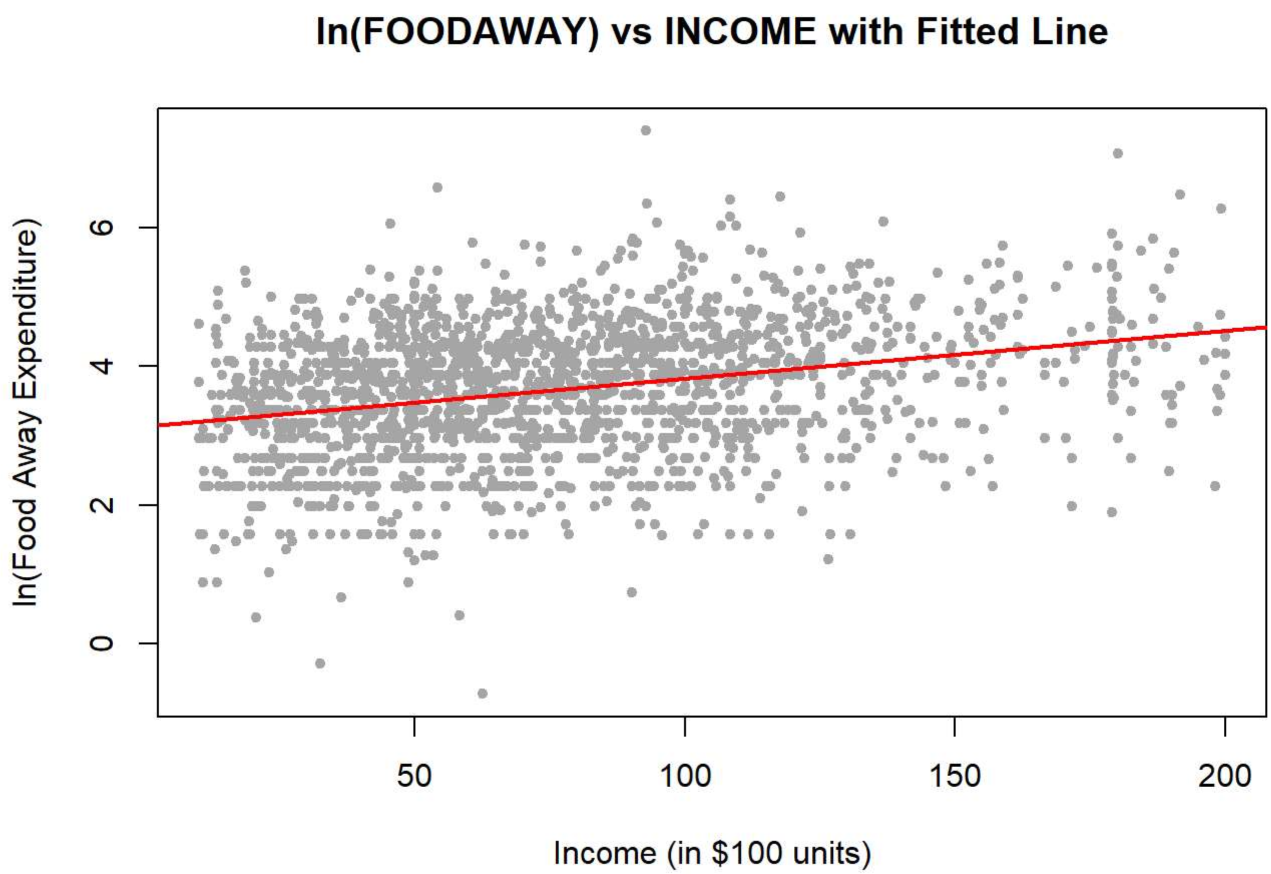
```
# c.
hist(log(cex5$foodaway[cex5$foodaway > 0]), main="Histogram of ln(FOODAWAY)")
```



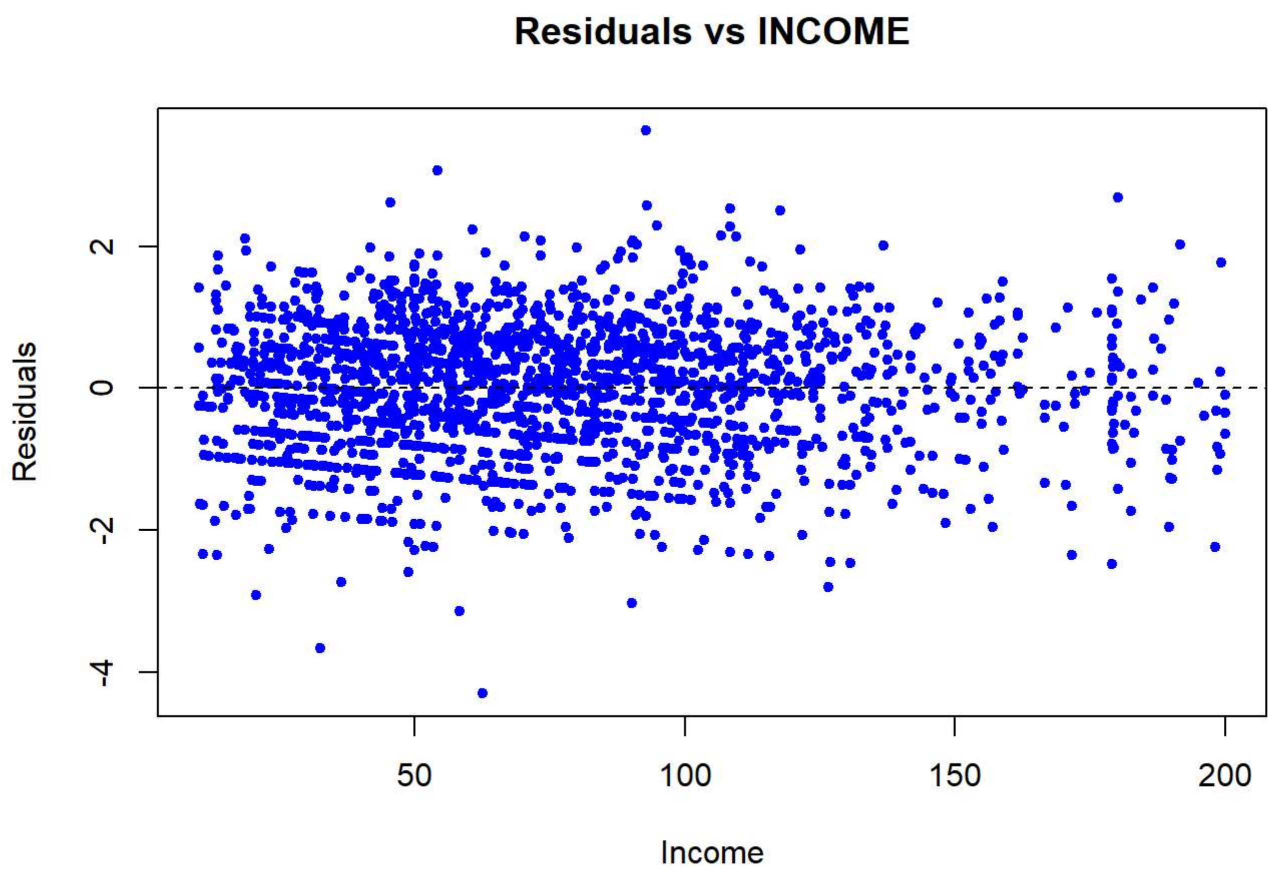
```
# d.
model_f <- lm(log(foodaway) ~ income, data = cex5[cex5$foodaway > 0, ])
summary(model_f)
```

```
##
## Call:
## lm(formula = log(foodaway) ~ income, data = cex5[cex5$foodaway >
##     0, ])
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.3034 -0.6003  0.0654  0.6119  3.6262
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 3.1364599  0.0420567   74.58   <2e-16 ***
## income      0.0069283  0.0004898   14.15   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.898 on 2003 degrees of freedom
## Multiple R-squared:  0.09083,    Adjusted R-squared:  0.09038
## F-statistic: 200.1 on 1 and 2003 DF,  p-value: < 2.2e-16
```

```
# e.
cex5_sub <- cex5[cex5$foodaway > 0, ]
model_d <- lm(log(foodaway) ~ income, data = cex5_sub)
plot(cex5_sub$income, log(cex5_sub$foodaway),
     main = "ln(FOODAWAY) vs INCOME with Fitted Line",
     xlab = "Income (in $100 units)",
     ylab = "ln(Food Away Expenditure)",
     pch = 20, col = "darkgrey")
abline(model_d, col = "red", lwd = 2)
```




```
# f.
e_hat <- residuals(model_d)
plot(cex5_sub$income, e_hat,
     main = "Residuals vs INCOME",
     xlab = "Income",
     ylab = "Residuals",
     pch = 20, col = "blue")
abline(h = 0, col = "black", lty = 2)
```


Residuals vs INCOME
Residuals
2
0
-2
-4
50
100
150
200
Income